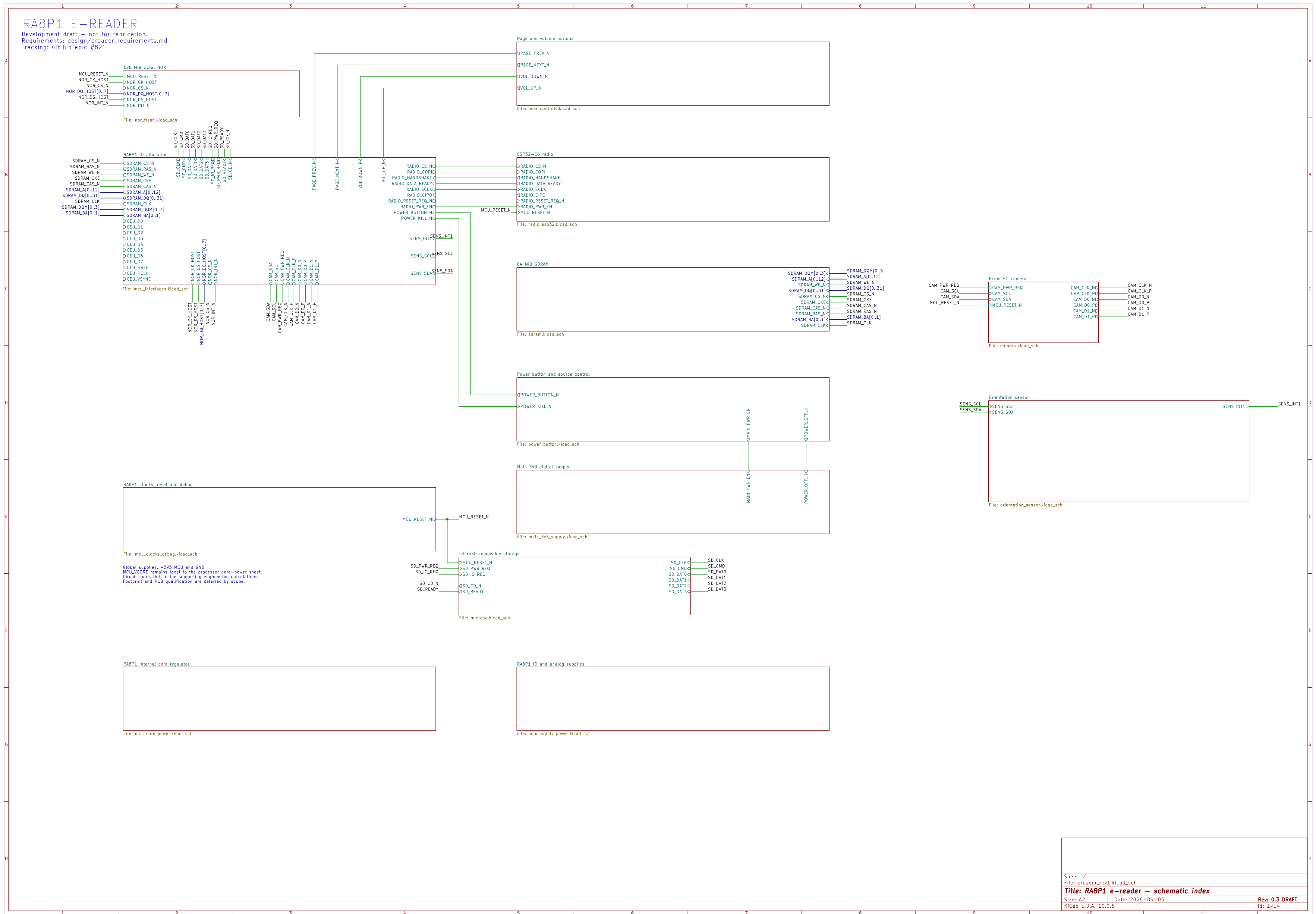


Development draft – not for fabrication.
Requirements: design/ereader_requirements.md
Tracking: GitHub epic #821.



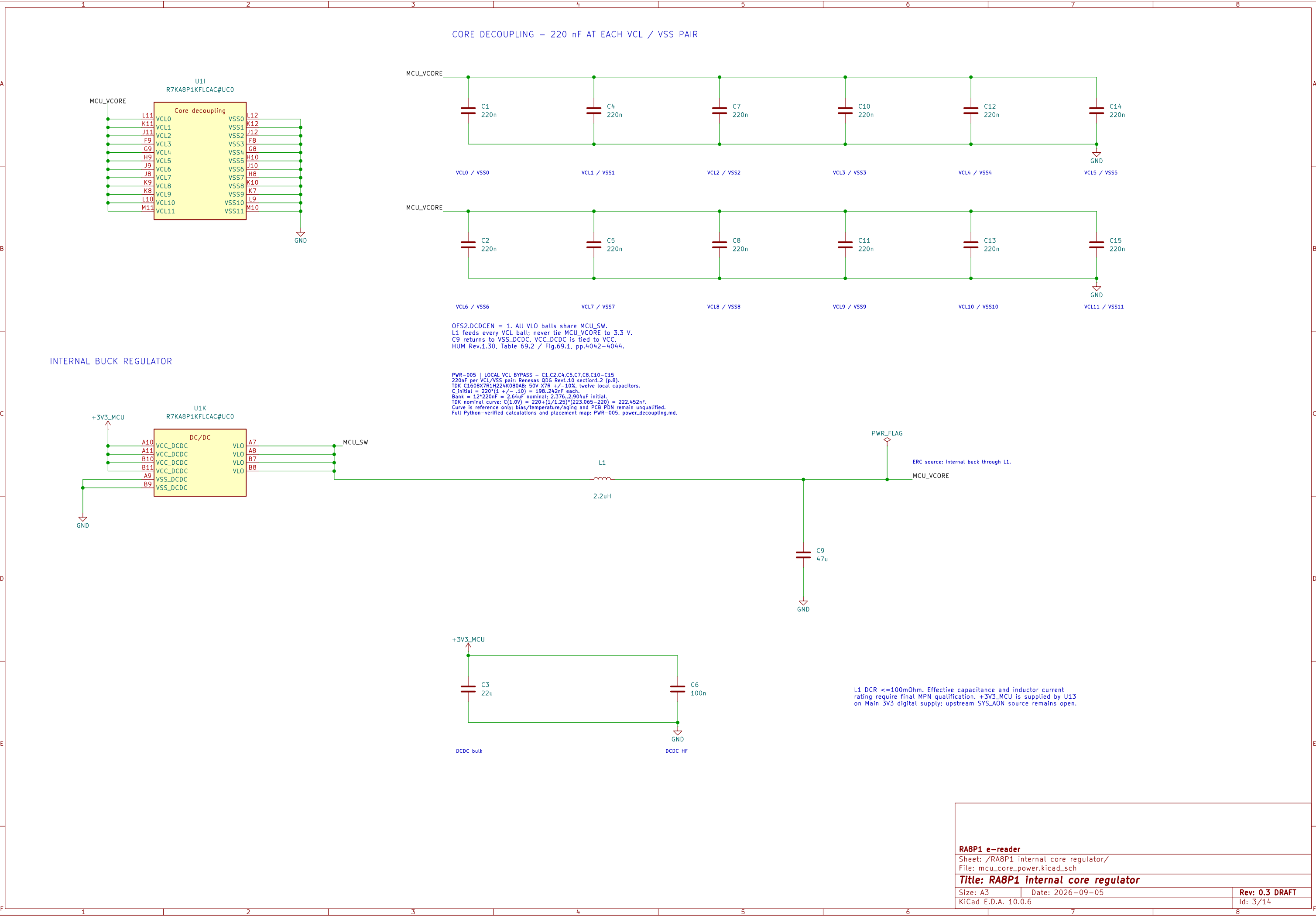
DRAFT: Existing processor units retained. Signal allocation and clocks/debug wiring are not yet complete.

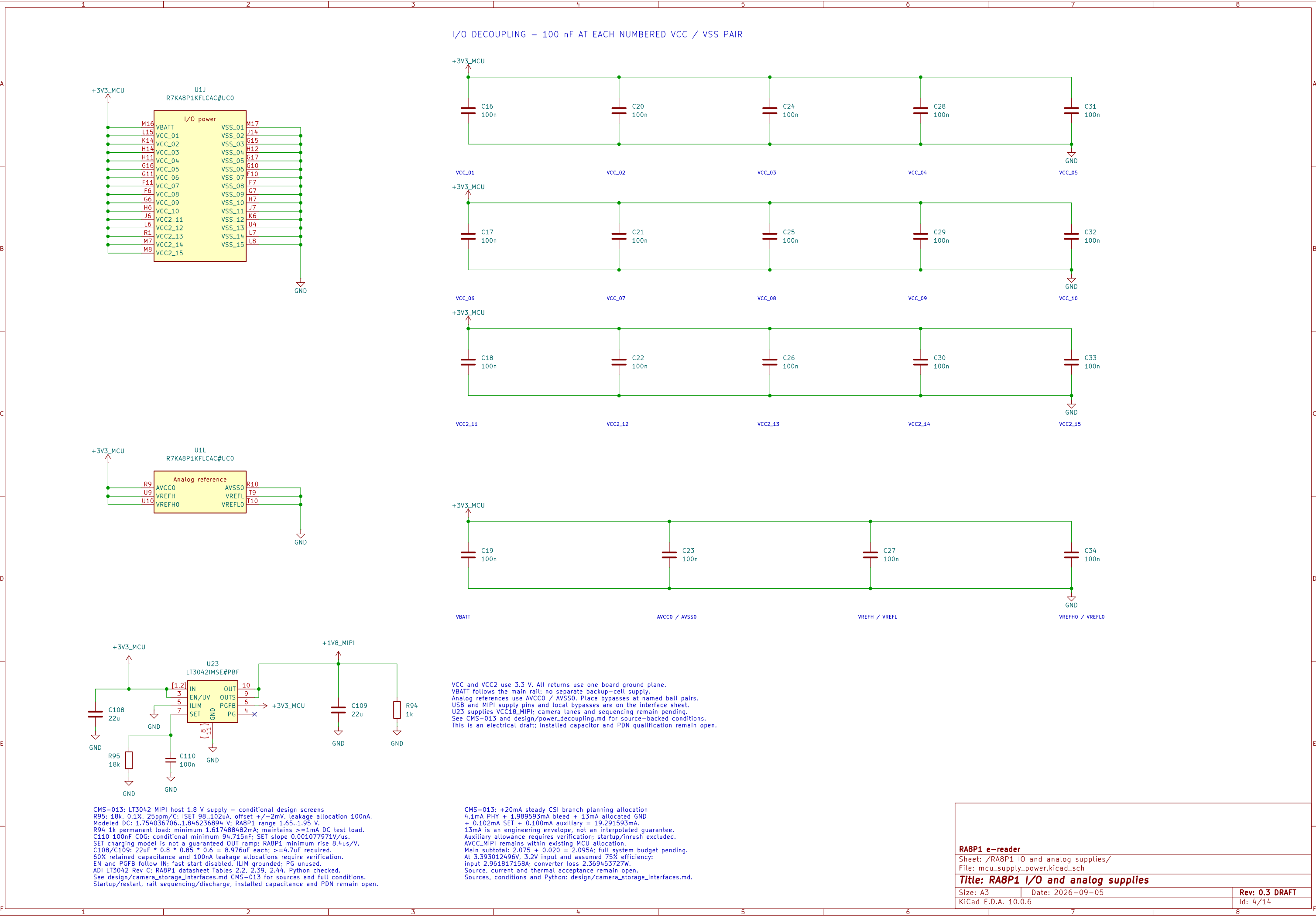
The diagram illustrates the IO allocation for the RABP1 e-reader. It includes several key components and their connections:

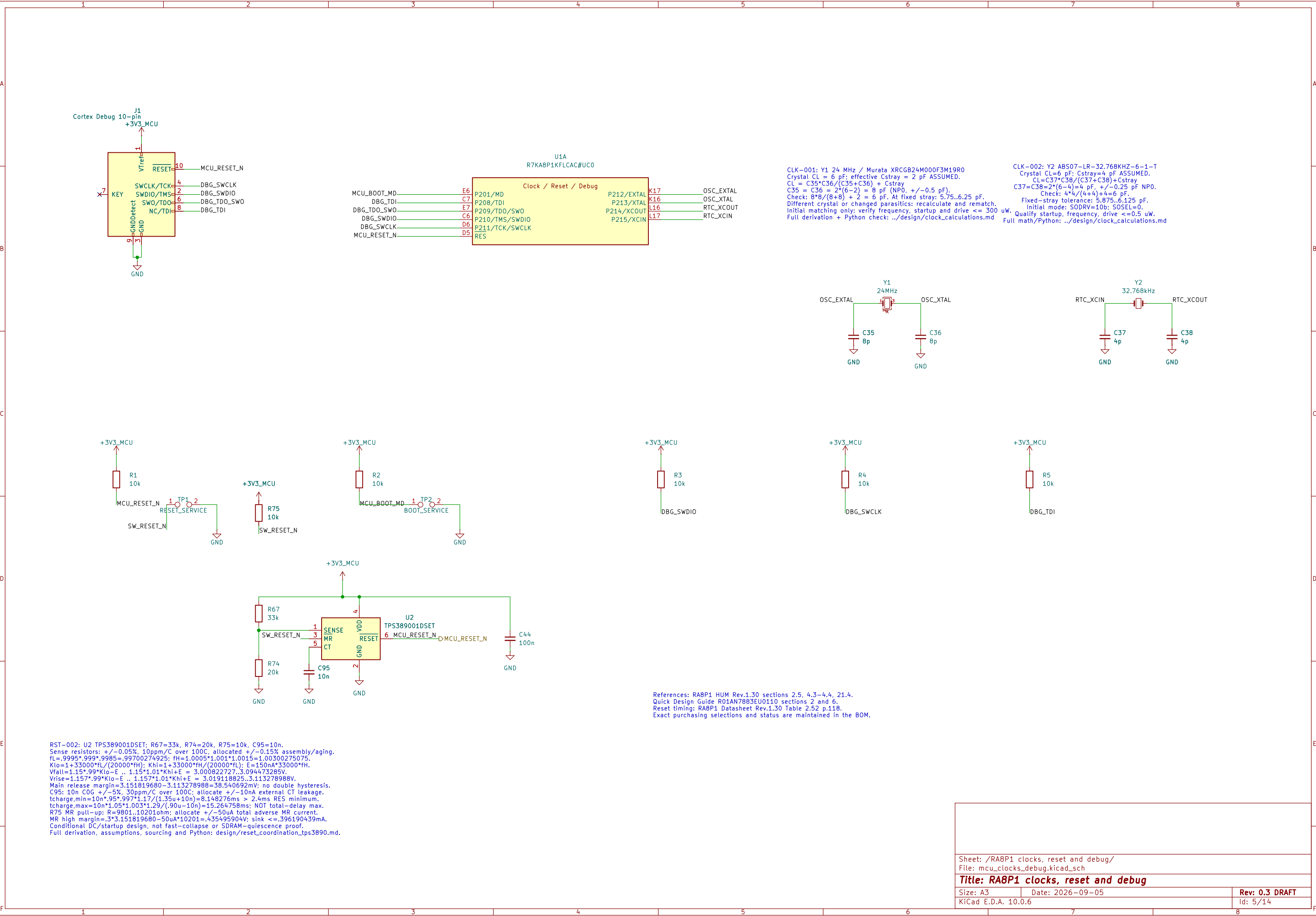
- Power and Grounding:** +3V3_MCU and +3V3_USBHS_A are connected to various components. A 120R resistor (FB1) is used for the +3V3_MCU supply. A 10uF capacitor (C63) is connected to the +3V3_USBHS_A supply.
- USB Interface:** The USB-004 module is connected to the +3V3_MCU and +3V3_USBHS_A supplies. It includes a 120 ohm resistor (FB1) and a 100 MHz capacitor (C63). The module is connected to the +3V3_MCU and +3V3_USBHS_A supplies.
- SDRAM:** The SDRAM is connected to the +3V3_MCU and +3V3_USBHS_A supplies. It includes a 120 ohm resistor (FB1) and a 100 MHz capacitor (C63). The module is connected to the +3V3_MCU and +3V3_USBHS_A supplies.
- GPIOs:** The GPIOs are connected to the +3V3_MCU and +3V3_USBHS_A supplies. They include a 120 ohm resistor (FB1) and a 100 MHz capacitor (C63). The module is connected to the +3V3_MCU and +3V3_USBHS_A supplies.
- Other Components:** The diagram also shows various other components like capacitors (C40, C41, C42), resistors (R43, R6), and modules (U1B, U1C, U1D, U1E, U1F, U1G, U1H, U1M, U1N, U1O, U1P, U1Q, U1R, U1S, U1T, U1U, U1V, U1W, U1X, U1Y, U1Z).

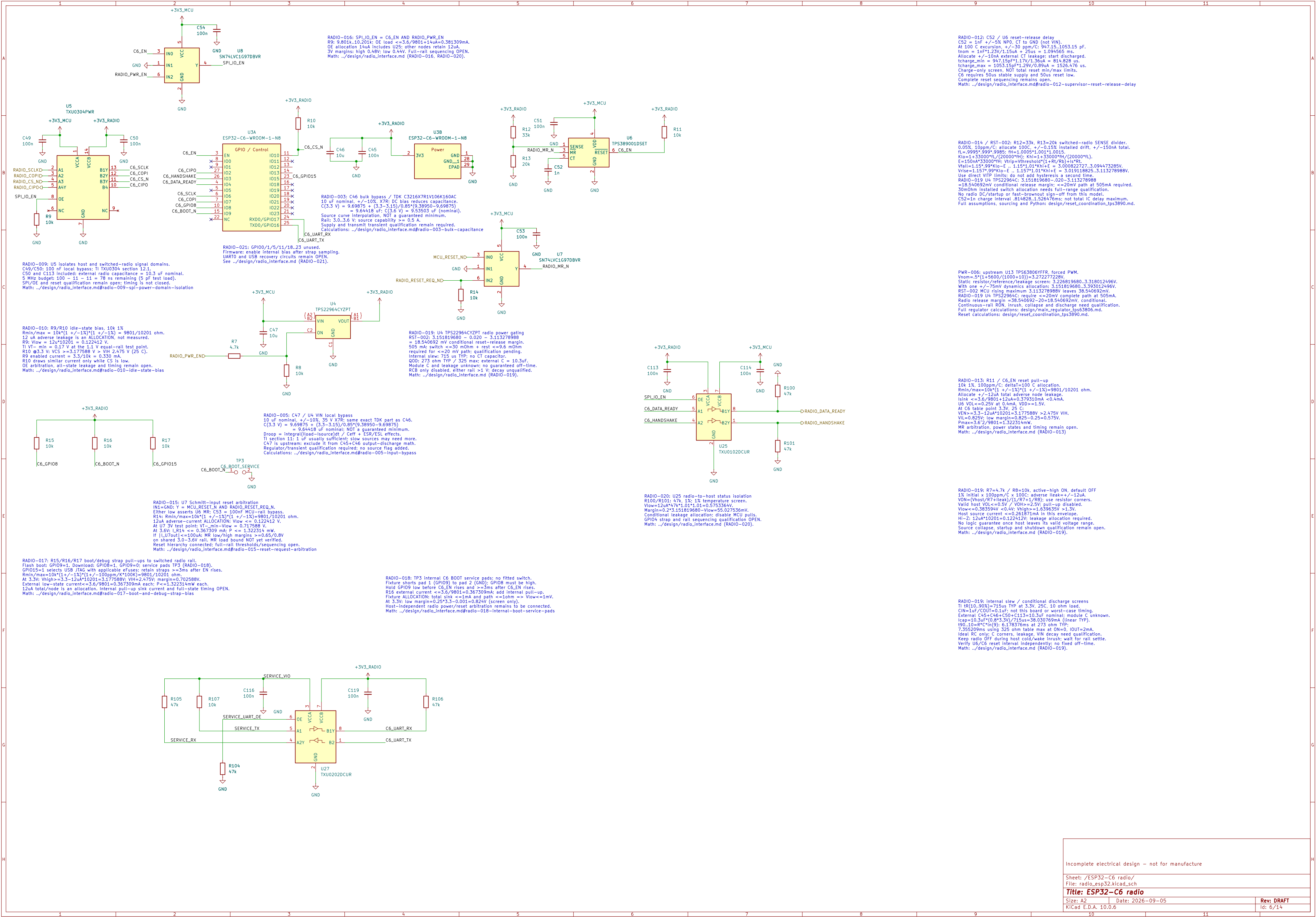
The diagram is a detailed schematic showing the electrical connections between various components. It includes a title block with the following information:

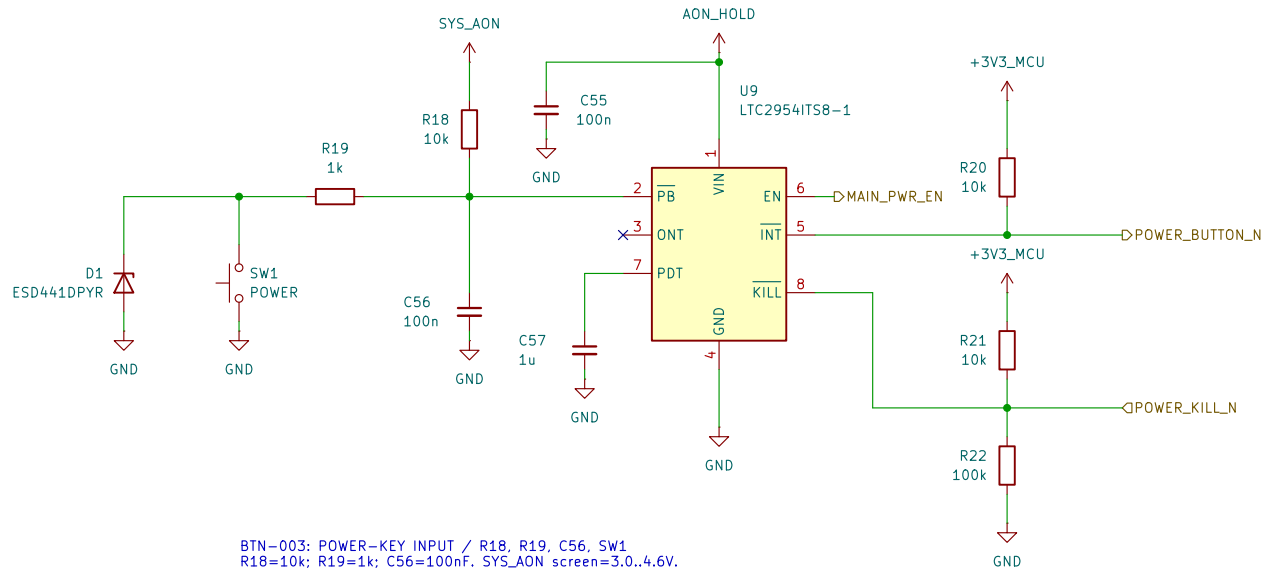
- Title:** RABP1 IO allocation
- Sheet:** /RABP1 IO allocation/
- File:** mcu_interfaces.kicad_sch
- Size:** A2
- Date:** 2026-09-05
- Rev:** 0.3 DRAFT
- Id:** 2/14





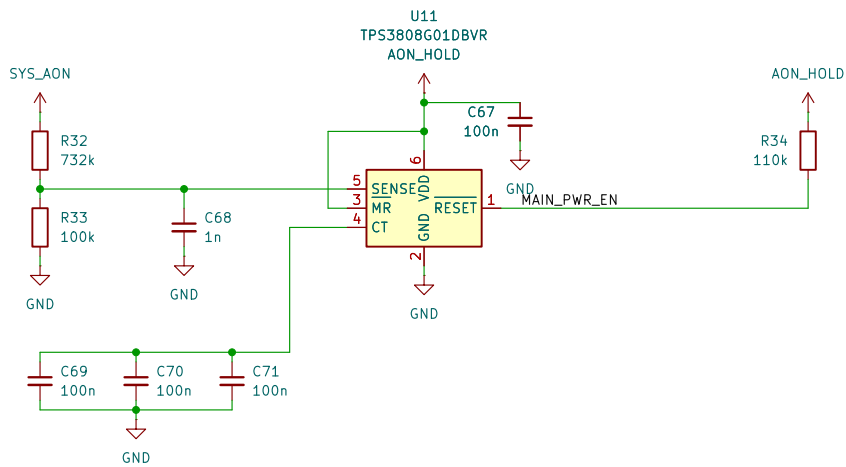






BTN-003: POWER-KEY INPUT / R18, R19, C56, SW1
R18=10k; R19=1k; C56=100nF. SYS_AON screen=3.0..4.6V.
R bounds: nominal*(0.99*0.99)..nominal*(1.01*1.01).
At PB=0.6V: Isource<=(4.6-0.6)/9801 + 15uA + 12uA
<435.13uA; Isink=0.6/(1020.1+0.5)>587.88uA.
Thus the closed switch pulls PB below its 0.6V falling threshold.
Min contact current=3/(10201+1020.1+0.5)-12uA>255.34uA.
Open contact>=2.87748V; SW1 minimum rating=10uA at 2V.
C56 discharge peak<=4.6/980.1<4.694mA < SW1 50mA rating.
12uA is a leakage ALLOCATION; ESD/transient validation required.
Full derivation + Python: ../design/single_button_power.md BTN-003/005/010.

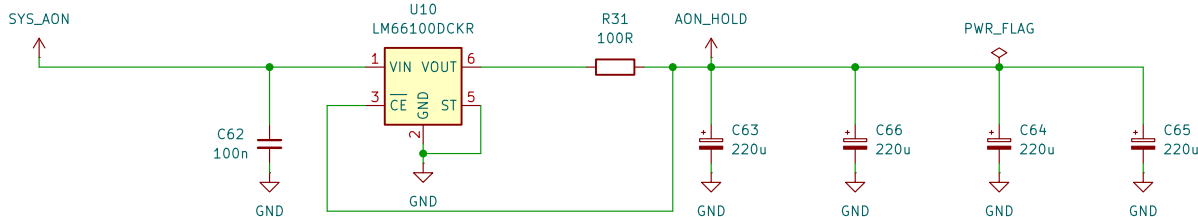
BTN-004: HARDWARE FORCED-OFF TIMING / C57
C57=1uF; TDK C2012X7R1E105K125AB, 25V X7R +/-10%.
C[uF]=1.56e-4*(t_added[ms]-1).
t_force,nom=64ms + 1ms + 1/1.56e-4=6475.256ms (6.48s).
C tolerance/temp only: 1*0.90*0.85=0.765uF.
1*1.10*1.15=1.265uF -> 4.969..8.174s using NOMINAL IC equation.
Not guaranteed timing: excludes IC spread, DC bias, aging, leakage.
ONT intentionally open: internal turn-on debounce=26..41ms.
Release the initial ON press; after forced off, release then press again.
Full calculation + Python: ../design/single_button_power.md BTN-004/005.



SYS-007 / SENS-010: SHUTDOWN / U12, Q1-Q3, R35-R40, R70-R71
POWER_OFF_H=NOT(MAIN_PWR_EN); held-domain VCC=2.7..4.6V.
Each gate branch: 1k series / 1M pulldown; 1% +100ppm/C, 100C.
4-gate peak<=4*4.6/(1000*0.99^2)=18.773595mA <50mA.
Includes future USB-clear; static<=22.773595uA <100uA.
VGSmin=(2.6-1020.1*1uA)/(1+1020.1/980100)=2.596278V.
1uA per gate/board leakage (is a qualification allocation).
RUN/KILL/discharge/USB-clear FET drains remain separate.
R39||R40 + Q2: Rmax=22*1.01^2/2+0.2=11.4211ohm.
Irc<=2.5mA: 1.5mA key-filter return +1mA other injection.
Vinf=Irc*Rmax=28.552750mV; Cmain<=1mF.
Total=10ms+Rmax*Cmain*ln((3.6-Vinf)/(0.09-Vinf))+10ms
=66.398757ms <75.794309ms hold; margin9.395552ms.
Each 22R: P<=3.6^2/(22*0.99^2)=0.601052W; 1W at70C.
Qualify FET resistance, response, derating, backfeed and cycling.
Gate/transient charge shares 1uC reserve; allocations need validation.
VBATT firmware/option invariant remains unimplemented.
Sources/Python: ../design/sensor_alternative.md SENS-010;
../design/system_power_design.md SYS-007 for topology.

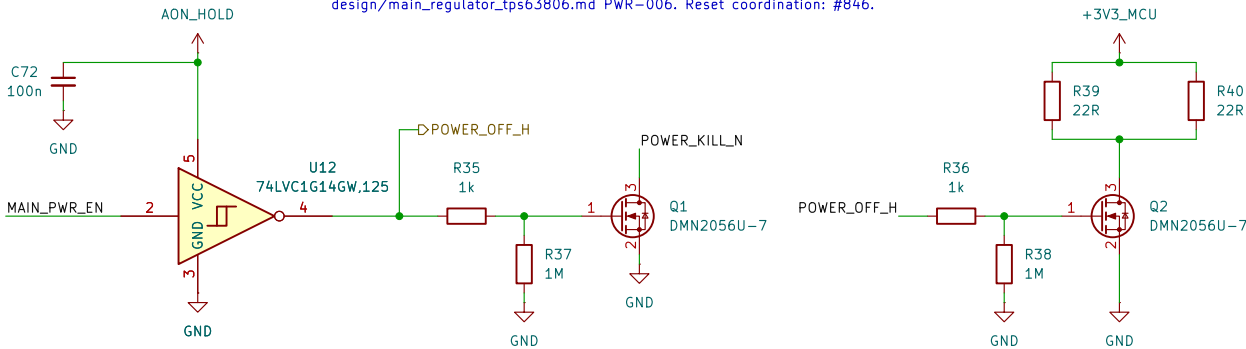
INCOMPLETE POWER CONTROL - NOT FOR FABRICATION
PB, INT, KILL, held supply and source supervisor are connected.
U12/Q1/Q2 provide fault KILL clamp and main-rail discharge.
MAIN_PWR_EN feeds raw U16; held Q3 separately clamps U13 EN.
SYS_AON source and USB permission-latch clear remain to be built.
Current converter: U13 TPS63806; design/main_regulator_tps63806.md PWR-006.
MCU reset: design/reset_coordination_tps3890.md RST-002.
Radio release margin and system shutdown coordination remain open (#846).

ERC supply path: U10 VOUT through R31.
PWR_FLAG declares that series-fed source only;
raw SYS_AON source remains unimplemented.



SYS-007 / SENS-010: HELD SUPPLY / U10, R31, C62-C66
C63-C66: 4 x 220uF/10V; Cmin=880uF*0.9^3=641.52uF.
VIN=raw SYS; VOUT->R31->reservoir; CE senses reservoir.
Icontrol<=1.825mA incl 1.100mA bank leakage screen.
Reverse allowance=0.817mA; Itotal<=2.642mA.
Vinitial=3.263705831-max(0.250,1.825mA*102.24ohm)
=3.013705831V; shared transition reserve=1uC.
thold=[641.52uF*(Vinitial-2.7V)-1uC]/2.642mA
=75.794309ms; conditional allocation.
Main C<=1mF. Rdis<=11.4211ohm, return/injection<=2.5mA;
10ms response +46.398757ms to90mV +10ms low hold
=66.398757ms total; hold margin=9.395552ms.
Recovery: raw>=3.70V, held>=3.25V; 1s settling PLUS
full source qualification; recharge model=0.657961s.
Qualify startup leakage, raw VIN=0, backfeed and cycling.
See ../design/sensor_alternative.md SENS-010;
../design/system_power_design.md SYS-007 for topology.

SYS-007 / PWR-006: SOURCE SUPERVISION / U11, R32-R34, C67-C71
Vtrip=Vref*(1+R32/R33)+Isense*R32.
Nominal=0.405*(1+732k/100k)=3.369600V.
Vref +/-2%, Isense +/-25nA; each R +/-0.1%, 25ppm/C, 100C.
Corner trip=3.263705831..3.476597632V.
C68: tau=(732k||100k)*1n=87.980769us (nominal).
C69-C71: 3x100n C0G; initial/temp=284.145..316.827nF.
CT delay=300/175+0.0005=1.714786s nominal.
0.6*(284.145/175+0.0005)=0.974511s MODEL, not guarantee.
Require qualified reset hold >=0.90s >650ms KILL blank +10ms.
R34=110k; ENhigh>=2.7-3.8uA*110k*1.01^2=2.273598V.
POR sink=(1.3-0.2)/(110k*0.99^2)+3.8uA=14.003041uA.
MAIN_PWR_EN drives raw U16 buffer; held Q3 clamps TPS63806 EN.
Held U12 also drives KILL/discharge; USB-clear still needed.
Current EN network and rail calculations: PWR-006.
Sources/Python: design/system_power_design.md SYS-007;
design/main_regulator_tps63806.md PWR-006. Reset coordination: #846.



Sheet: /Power button and source control/
File: power_button.kicad_sch

Title: RABP1 E-reader - power button and source control

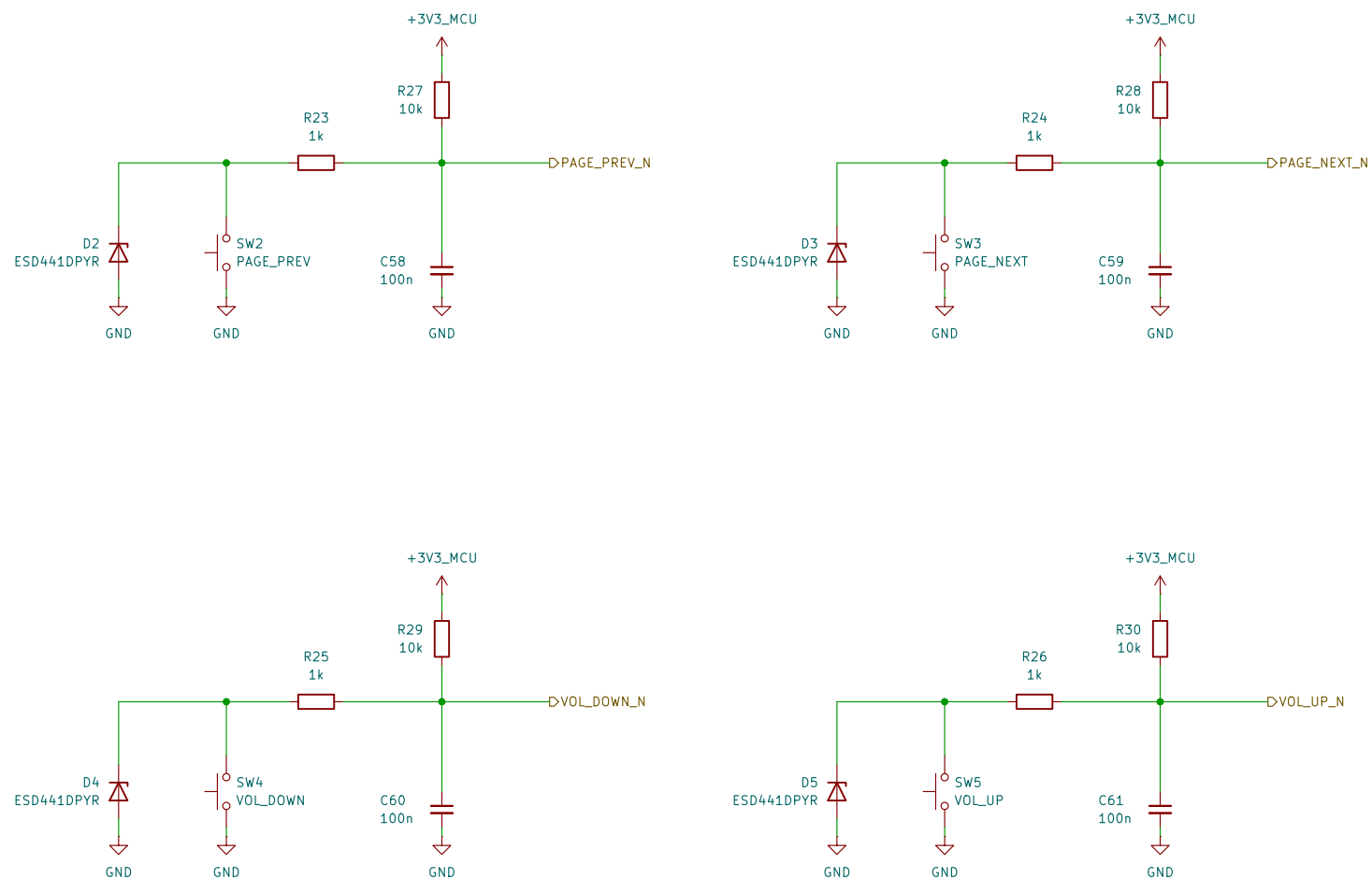
Size: A3

Date: 2026-09-07

Rev: 0.3 DRAFT

KiCad E.D.A. 10.0.6

Id: 7/14



CMS-009 - R26 INDEPENDENT PAGE / VOLUME KEYS
SW2, SW5: R23, R26 to R1, R27, R30, 10k; C56, C61=100n.
D2, D3: 5404, 04PYR 01, R27, R30, 10k; C62D2, before the 10k resistors.
VCC+3V3, MCU: 3.0, 3.6V screening envelope.
Resistors: $\pm 1\%$, plus $\pm 100\text{ppm}/^\circ\text{C}$ over 100C.
RPU=10k $\times(0.99^{(2-1.012)}=9801.1\pm 2021$ ohm.
RS=1k $\times(0.99^{(2-1.012)}=980.1\pm 202.1$ ohm.
Contact ≤ 0.5 ohm; adverse leakage allocation=2.2uA/key;
1uA MCU +0.1uA TVS +0.1uA capacitor +1uA board.
Capacitor and board leakage terms are design allowances.

$V_{LOW} = (V_{CC}/R_{PU} + I_{leak}) / (1/R_{PU} + 1/(R_S + R_{contact}))$.
 Worst case $V_{LOW} = 0.342V$; correlated VIL margin $> 0.315V$.
 VHIG $= 3 - 2uA \cdot 10201 = 2.9775578V$; VIH margin $> 0.57755V$.
 $I_{contact} = 3 / (10201 + 1020.6) - 2uA > 265.14uA$.
 Open contact $= 2.97745V$; switch minimum load $= 10uA$ at $2V$.
 Peak capacitor discharge $= 3.6/980.1 = 3.674mA$ ($< 50mA$).
 Four keys held: $4 \cdot (3.6 / (980.1 + 980.1) + 2uA) < 1.345mA$.

Full derivation, corner enumeration and verified Python:
[../design/camera_storage_interfaces.md](#) CMS-009.

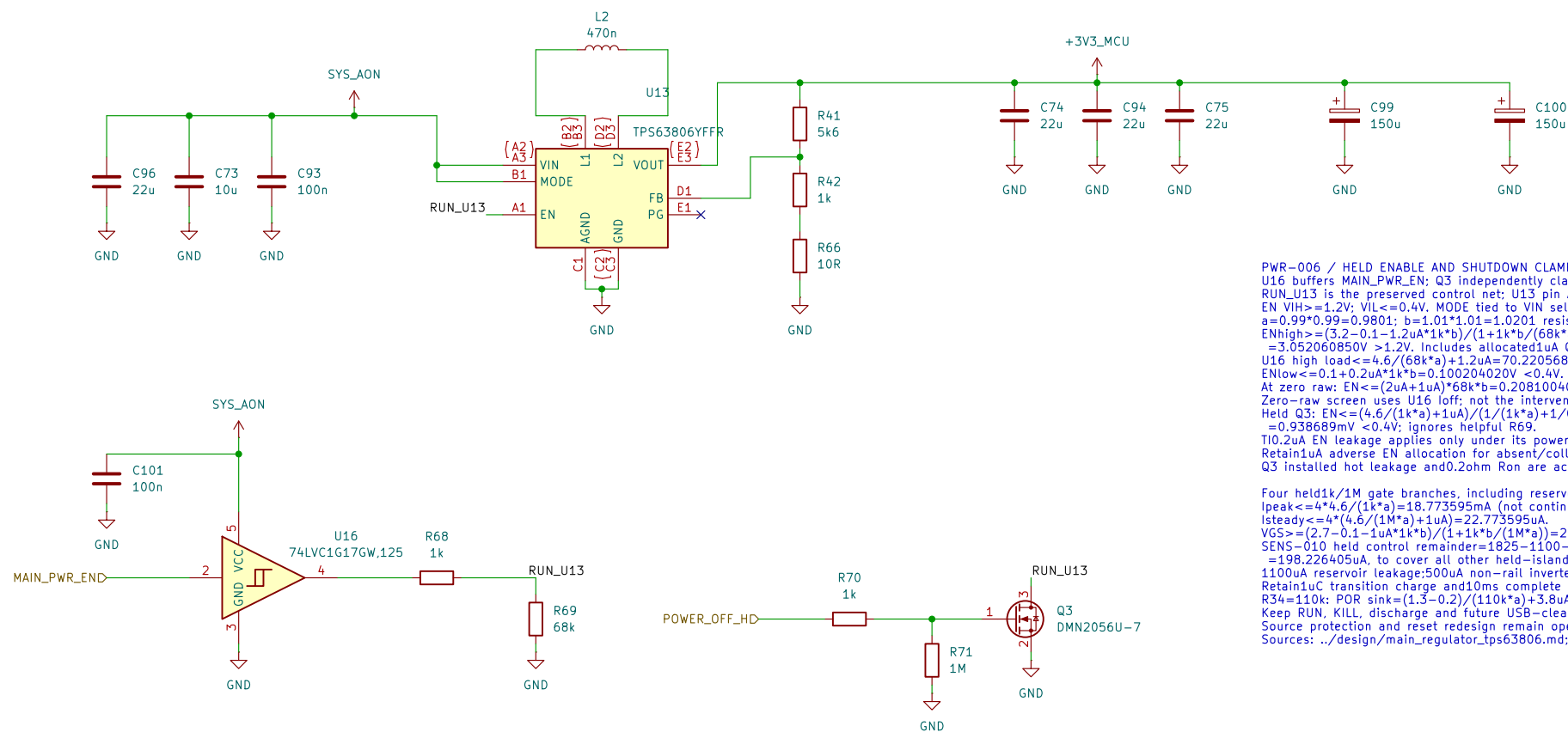
CMS-009 – FILTER TIMING AND BUTTON BEHAVIOR
C58..C61: TDK C1608X7R1H104K080AA, 100nF, X7R, 50V.
Tolerance / temperature-only capacitance bounds:
 $C_{min}=100nF \times 0.90 \times 0.85=76.5nF$; $C_{max}=100nF \times 1.10 \times 1.15=126.5nF$.
DC bias and aging excluded; quality effective capacitance.

Nominal $\tau_{\text{assert}} = (10k || 1k) * 100n = 0.090909ms$.
 $\tau_{\text{assert}} = \tau_{\text{assert}} * \ln((0.2 - 1/11) / (1 - 1/11)) = 0.193ms$.
 $\tau_{\text{release}} = -10k * 100n * \ln((1 - 0.8) / (1 - 1/11)) = 1.514ms$.
 RC filters noise; it does NOT provide complete debounce.
 Require 20ms continuously stable state; switch bounce = 10ms.
 IRQ starts wake/debounce, not an immediate page/volume action.

Opposing page or volume keys cancel their associated action. Ignore boot—held ordinary actions until release; no auto—repeat from a stuck key. Disable MCU internal pulls. In hard-off, these four keys cannot start power. The separate POWER switch controls hardware start/forced-off. ESD441 is not a 3.3V clamp; residual MCU voltage, series resistor pulse stress and ESD current return need physical tests.

Full corner calculations, Python and wake-register contract:
 ../design/camera_storage_interfaces.md CMS-009.

MAIN DIGITAL SUPPLY: TPS63806 / PWR-006
3.272277 V nominal; forced PWM; held shutdown clamp.
Divider and bypass calculations: design/main_regulator_tps63806.md
Reset coordination (#846), source protection and qualification remain open.
DEVELOPMENT SCHEMATIC – NOT FOR POWER-UP OR FABRICATION.



```

PWR-006 / HLD ENABLE AND SHUTDOWN CLAMP
U16 buffers MAIN_PWD_EN; Q3 independently clamps EN during raw collapse.
RUN_U13 is the preserved control net; U13 pin A1 is now EN.
EN VIH>=1.2V; VIL<=0.4V. MODE tied to VIN selects forced PWM.
=0.99+0.99-0.9801; b=1.01*1.01+1.0201 (resistor factors)
= (3.2-0.1-1.2uA)*1k*b/(1.0201+1.0201)
= 3.052060850uA. Included allocationU13 off leakage.
U16 high load = 4.6/(68k)+1.2uA=70.220568uA <100uA.
ENlow = 0.1+0.2uA*1k*b = 0.100204020V <0.4V.
At zero raw: EN<=(2uA+1uA)*68k*b = 0.208100400V.
Zero-raw screen uses U16 loff; not the intervening undefined region.
Held Q3: EN<= (4.6/(1k)+1uA)/(1/(1k)+1/0.2ohm)
= 0.9386mV <0.4V; ignores held U16.
T10.2uA EN leakage applies only under its powered test conditions.
RetainU1A adverse EN allocation for absent/collapsing raw.
Q3 installed hot leakage and0.2ohm Rn are acceptance allocations.

```

```

Four held1k/1M gate branches, including reserved USB-clear:
Ipeak<=4*(16/(1k^a))=18.773595mA (not continuous ratings).
Isteady<=4*(6/(1M^a+1A))=22.773595mA.
VG5=(2.7-0.1)/(1mA*1k^b)+((1+1k^b)/(1M^a))=2.506278V.
SENS=high-level control, remainder=862.5mA/22.773595
=198.226450uA, to cover all other held-island loads.
1100uA reservoir leakage:500uA non-rail inverter-input allocation.
RetainLCU transition charge and10ms complete response allocation.
R34=110k: POR sink=(1.3-0.2)/(110k^a)+3.8uA=14.003041uA.
KILL=kill switch, no charge and no current source separate.
Source protection and reset redesign remain open; no power-up approval.
Sources...:/design/main_regulator_tps63806.mt; SENS=010 revision.

```

PWR-Vom / FEEDBACK AND LOCAL PASSIVES
 $V_{nom} = 5.00 \times (1 + 5600 / (1000 + 10)) = 3.27227228 \text{ V}$
R41: 0.02%, 5ppm/C; R42: 0.01%, 2ppm/C; R66: 1%, 200ppm/C.
Installed boards: initial "TCR(100C)"aging"assembly, then ohms.
 $R41 = 5600 \times (1 + (-0.0002)) \times (1 + (-0.0005)) \times (1 + (-0.001))$
 $\times (1 + (-0.0005)) \div (-0.02) = 5587.6629237, 5612.349243 \text{ ohm}$
 $R42 = 1000 \times (1 + (-0.0001)) \times (1 + (-0.0002)) \times (1 + (-0.001))$
 $\times (1 + (-0.0005)) \div (-0.01) = 998.190970, 1001.810970 \text{ ohm}$
 $R66 = 10 \times (1 + (-0.01)) \times (1 + (-0.02)) \times (1 + (-0.01)) \div (-0.05)$
 $= 9.954980, 10.455020 \text{ ohm}$ one combined aging/assembly 1%.
 $V_{in} = 5.00 \times (1 + 14160 / (R41 \times R42 \times R66)) = 5.00 \times (1 + 14160 / 1000 \times 1000 \times 10)$
 $V_{in} = 5.005 \times (1 + 14160 / (R42 \times R66 \times 10)) = 5.005 \times (1 + 14160 / (1000 \times 1000 \times 10))$
 $V_{static} = 3.226819680, 3.318012496 \text{ V (FPWM only)}$
One combined + / -75mV regulation/ripple/transient/routing allowance:
 $V_{complete} = 3.315189680, 3.393012496 \text{ V}$: qualification required.
SDRAM static VOH margin = $2.4 - 0.73 = 3.93012496 \text{ V} = 2.4891253 \text{ mV}$.
This is not measured S/I/noise headroom or reset qualification.

12: 470nH +/-20%; catalog spread alone is insufficient.
Screen incoming0.412..0.518uH; allocated operating+/-10%
=+/-3708..0.5698uH versus T137..0.57uH. Quality hot/current.
C74/C75/C94: *322uF*0.8*0.85*0.6=6.2928uF > T121uF
C96: 22uF*0.8*0.85*0.6=8.976uF > T14uF input minimum.
The capacitor combined resistance is guaranteed.
Samsung typical output bias retention at3.393012496V-90.221554%.
Input bias, aging, mounted impedance and startup need verification.
Full assumptions, sources and executable Python: click this note.

PWR-006 / LDA, STORAGE AND SHUTDOWN
Allocation: 1.80A operating; 2.25A cold start screen, not measured load.
Pwr: 3.393012496e+1, 8=6.107424332
Thermal: 2.544759372A, 2=2.544759372A
Losses: 0.678602499W, 2=0.678602499W
TJ90C efficiency: loss=0.678602499W
TJ90C=70C+78.8C/0.678602499W=123.474C.
2.544759372A is the final PCB thermal resistance.
TJ90C=70C+78.8C/3.393012496e+1=120.179037499W.
Converter limit, source limit, thermal and thermal envelope need validation.

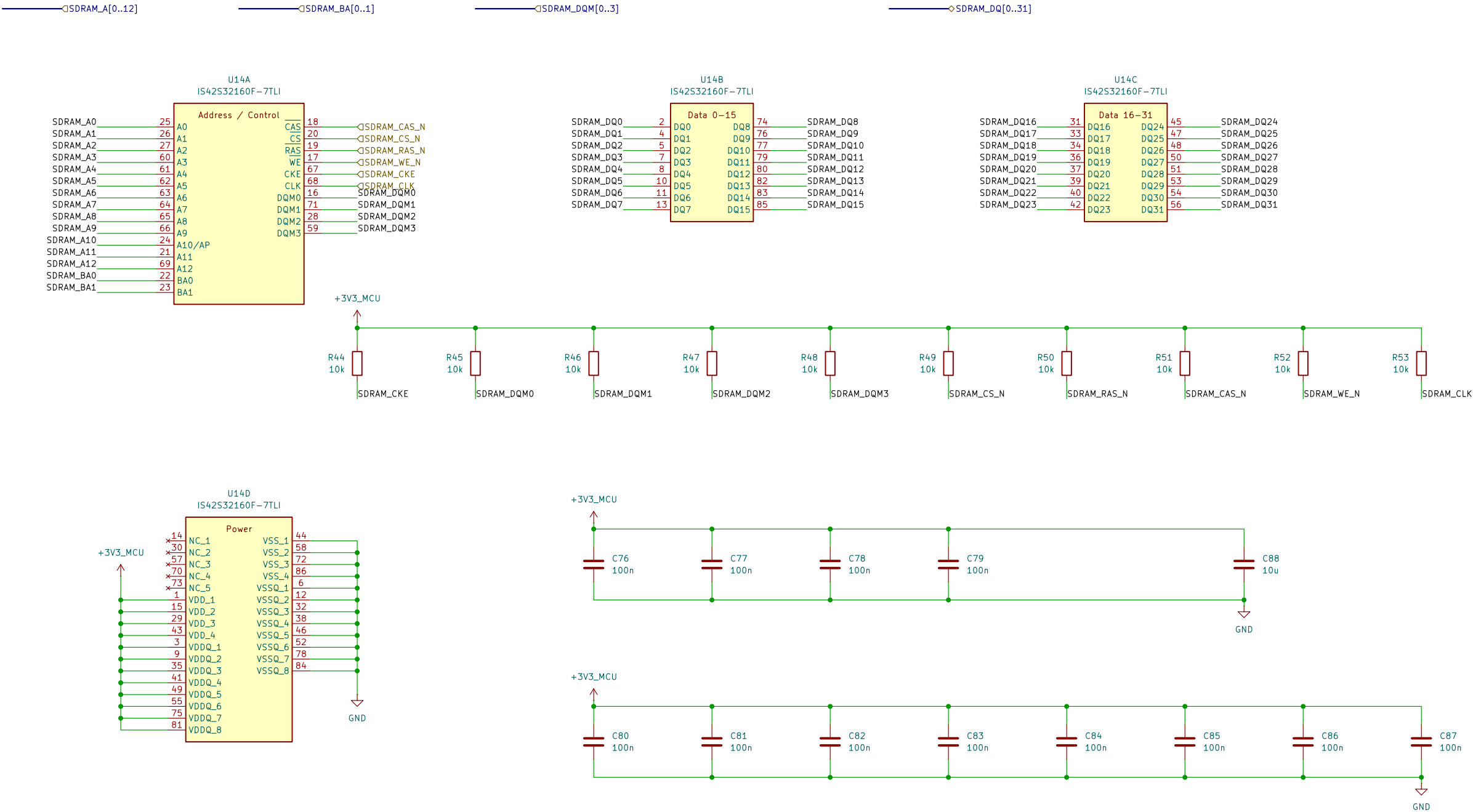
PWR=006 storage checkpoint (nominal, not effective minimum):
 Direct main469.21uF + C43 through FB1 10uF + key filters0.4uF
 =479.61uF cold; radio-on add±0.2uF =>489.81uF.
 C99/C100 remain±150uF each. C94 adds22uF to the output.
 Old bootstrap/private VCC/VyF capacitors were not main-rail storage.
 Whole-main ceiling remains1mF; reconcile later peripheral additions.
 Cout lower acceptance and distributed PDN/startup are separate checks.

```

SENS-010 discharge revision: Rmax1+0.42110hm: return/injection<=+2.5mA.
Includes1.5mA key return +1mA other backfeed: Vinf=0.028552750V.
t=1mF+11.42110hm*ln((3.6-Vinf)/(0.09-Vinf))+10ms+10ms
=66.3987575ms total, below475.794309ms screen.
SENS-010ms delay: Rmax1+0.42110hm: return/injection<=+2.5mA.
Held budget:1.825mA control +0.817mA reverse:1uC transition reserve.
198.226405uA control remainder must cover other held-island loads.
SYS_AON source protection, loaded startup and fast faults remain open.
RSN-002 / RADIO-019: release margin remains conditional on rail/path bounds.
No more power for the sensor and control and connection are complete.
Full assumptions: ../design/main_regulator_tps63806.mt
shutdown revision/Python: ../design/sensor_alternative.md SENS-010.

```

64 MiB SDRAM | ISSI IS42S32160F-7TLI
Signal and reset—default circuits | CMS-010.
Startup, SI/timing and hardware qualification remain open.



CMS-010 | SDRAM power and bypass
VDD and VDDQ share +3V3_MCU (3.0 to 3.6 V).
C76-C79: 4 x 100 nF for the four VDD pins.
C80-C87: 8 x 100 nF for the eight VDDQ pins.
C88: 10 uF local bulk; Cnom = 12 x 0.1 + 10 = 11.2 uF.
Included in the PWR-003 1 mF whole-rail limit.
Effective C, placement and PDN impedance require qualification.

CMS-010 | R44-R53: 10k reset defaults
Rail 3.0..3.6V: +/-1%, 100ppm/C. |dT| <=100C.
Rmin=10k*0.99*0.99=9801 ohm.
Rmax=10k*1.01*1.01=10201 ohm.
Ioff=5uA SDRAM+1uA MCU+1uA board=7uA.
VHIGHmin=3.0-7uA*10201=2.928593V (>2.0V VIH).
Board leakage and temperature screen require qualification.

CMS-010 | Pull load and power states
Sink leakage: 5uA SDRAM+1uA board=6uA.
Isink=3.6/9801+6uA=0.373309mA (<0.374mA).
Pmax=3.6^2/9801=1.322314mW per pull.
All ten LOW: 10*3.6/9801=3.673095mA rail:
10*3.6^2/9801=13.223140mW resistor heat.
Reserve 4mA within the 250mA SDRAM allocation.
CKE/DQM and CS_N default HIGH; CLK HIGH is not a clock.
Switched-rail pulls; self-refresh actively holds CKE LOW.
Startup, dynamic I/O, SI and shutdown qualification OPEN.

Sheet: /64 MiB SDRAM/
File: sdram.kicad_sch

Title: RA8P1 E-reader - 64 MiB SDRAM

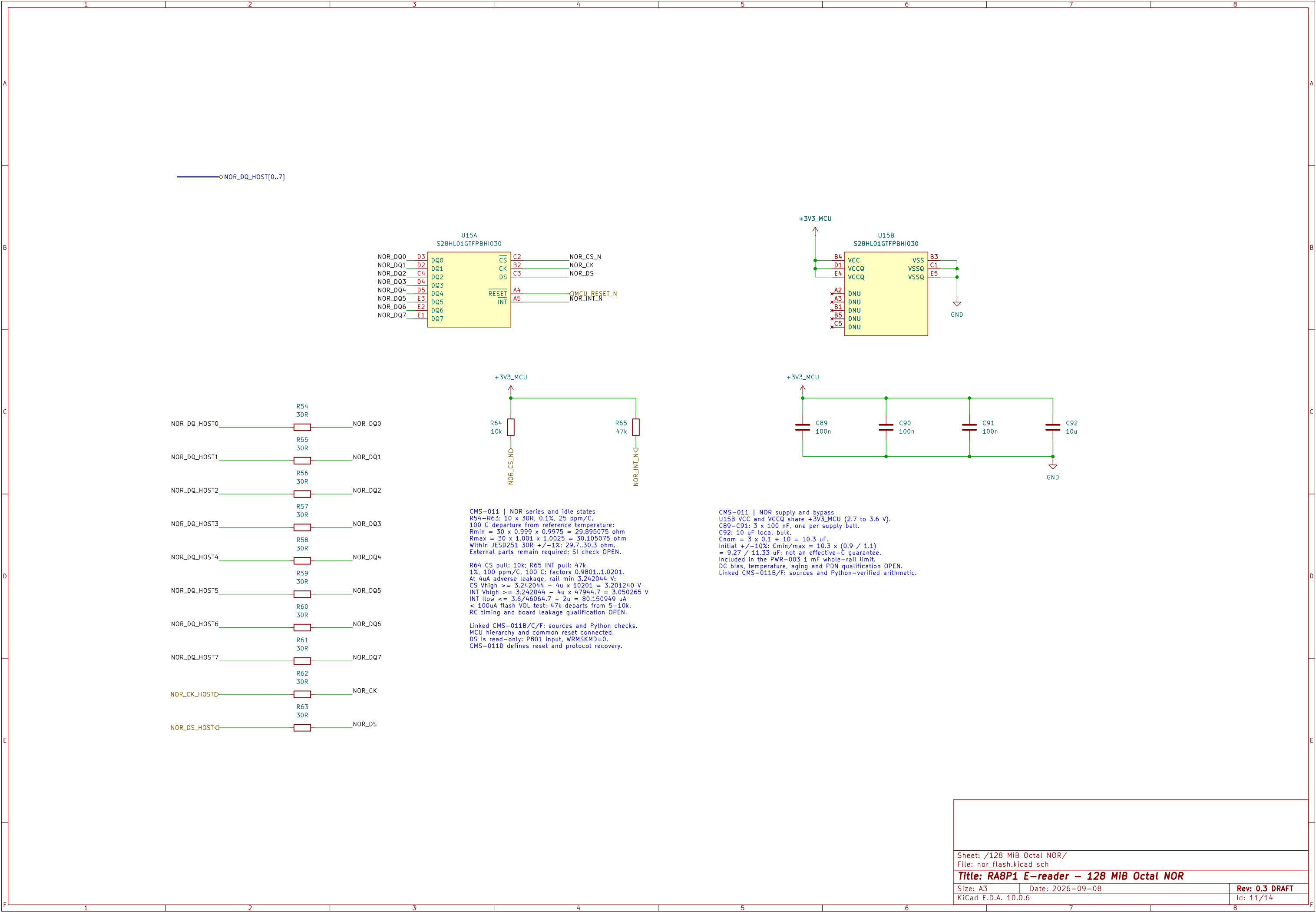
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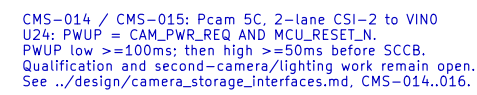
Date: 2026-09-08

Rev: 0.3 DRAFT

KiCad E.D.A. 10.0.6

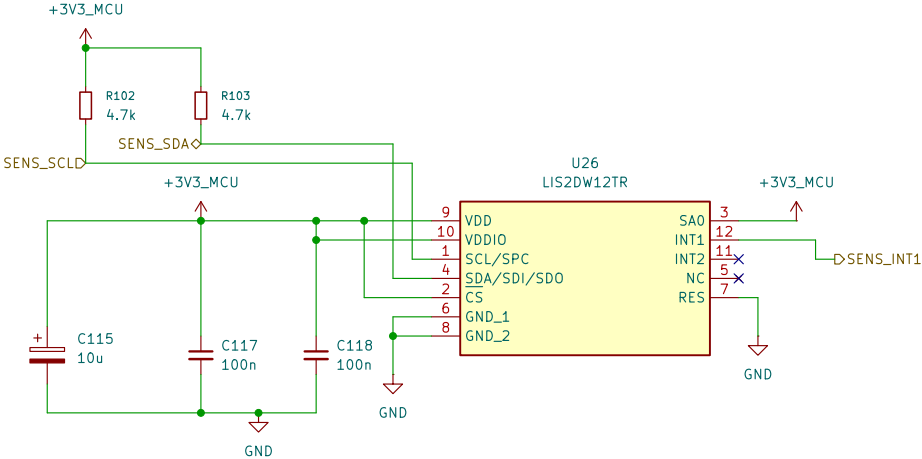
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Sheet: /Pcam 5C camera/ File: camera.kicad_sch		
Title:		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.6		Id: 13/14

ORIENTATION SENSOR – WIP
U26: LIS2DW12TR; ST DS11811 Rev 9 pin map.
VDD pin 9 / VDDIO pin 10: +3V3_MCU; CS / SA0 high for I2C address 0x19.
C117/C118: local 100n bypass; GND pins 6/8 and RES pin 7 grounded.
C115: 10u/16V aluminum EEE–FM1C100R; place near VDD pin 9 with C117.
R102/R103: 4.7k SCL/SDA pull–ups; obsolete ADXL VREG network removed.
Host connected: SCL P410/K13; SDA P409/R16; INT1 P306/D11.
Wait 20ms after supply valid. Power–cycle below 100mV for >=10ms.
Pending: sourcing fields, pin modes, timing/power/passive qualification.
See design/sensor_alternative.md; circuit qualification remains open.



Sheet: /Orientation sensor/
File: orientation_sensor.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 10.0.6

Rev:

Id: 14/14